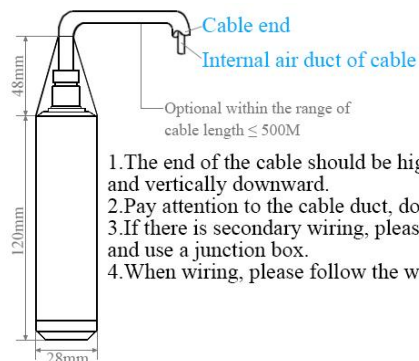


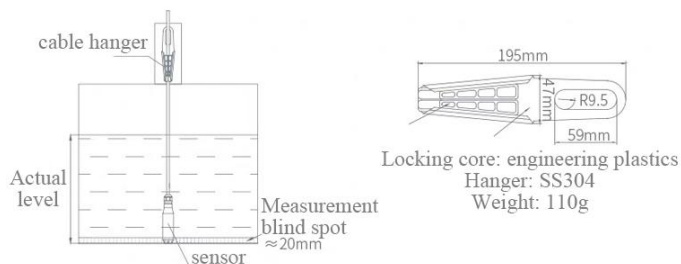
Product Size



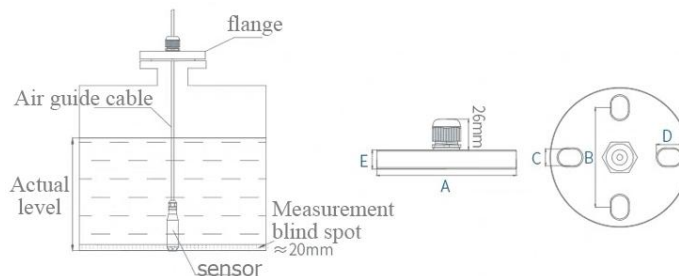
1. The end of the cable should be higher than the water surface and vertically downward.
2. Pay attention to the cable duct, do not let water enter or block it.
3. If there is secondary wiring, please take waterproof measures and use a junction box.
4. When wiring, please follow the wiring diagram.

Installation Precautions

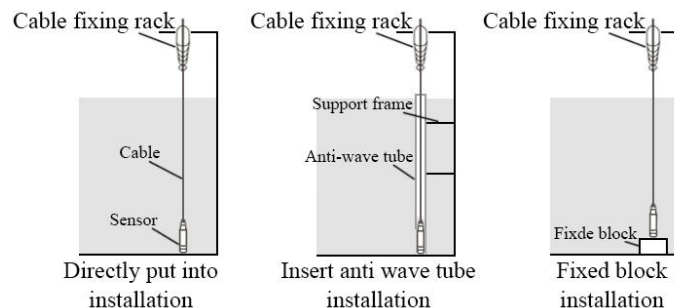
1. When installing, choose a place that is easy to operate and maintain. Avoid water inlet or mixing port.
2. It should be installed as far away from vibration and heat sources as possible.
3. There is an isolation diaphragm inside the pressure inlet of the sensor, so do not collide with foreign objects.
4. The probe needs to be regularly cleaned of silt to avoid clogging the inlet.
5. When installing the liquid level transmitter, the metal probe needs to sink to the bottom of the container.
6. The cable is waterproof, and wear, punctures, and scratches should be avoided during installation and use.
7. The cable length should be greater than the liquid level height and should be vertically downward to avoid water entering the cable duct and causing damage to the sensor.
8. Cable hangers can be used for installation, but please note that the cable length must be greater than the liquid level height. (As shown in the figure below)



9. Flange fixed installation, but please note that the cable length must be greater than the liquid level height. (As shown in the following figure)



Installation Matters



a. Directly put into installation, suitable for static liquids.

This installation is relatively simple, simply place the liquid level transmitter at the bottom of the liquid to be measured (leaving a small gap at the bottom of the sensor), and fix the cable with a cable fixing bracket. This installation location needs to be away from the liquid inlet and outlet agitator.

b. Installation is carried out by inserting anti wave tubes, suitable for liquids with high impurities or fluidity, such as sewage tanks.

Insert a steel pipe or PVC pipe with an inner diameter of about 45cm into the tested liquid, fix it vertically in water, make several small holes at different heights to allow water to flow smoothly into the pipeline, and then place the sensor component into the pipeline for use. In order to filter impurities such as sediment and eliminate the influence of dynamic pressure and fluctuations on the liquid level transmitter. The installation direction of the liquid level transmitter should be vertical.

c. Fixed block installation, suitable for high fluidity liquids.

Install a hollow fixed block with holes at the bottom of the liquid to secure the level transmitter and prevent it from being pushed too far by water, ensuring the vertical position of the transmitter and reducing the impact of waves on the sensor.

Liquid level transmitter



NOTICE

- Please check the product label to see if it is the same as your ordering parameters.
- Please read this instruction manual carefully before use. If you have any questions, please contact us for assistance.
- This product cannot measure high temperatures and corrosive liquids.
- When selecting cables, the cable length needs to be greater than the liquid level height.
- When installing the liquid level transmitter, the metal probe needs to sink to the bottom of the container.
- The cable is waterproof, and wear, punctures, and scratches should be avoided during installation and use.
- The probe of the liquid level transmitter should be regularly cleaned of silt to avoid blocking the pressure hole.
- There is an isolation diaphragm inside the pressure inlet of the sensor, so do not collide with foreign objects.

Description

Using an imported diffusion silicon piezoresistive pressure sensor as the measuring element, the precise measurement of liquid level depth is directly proportional to the static pressure of the liquid, and is converted into a standard (current or voltage) signal output through a signal conditioning circuit, establishing a linear correspondence between the output signal and the liquid depth to achieve liquid depth measurement.

Used to measure the height of non corrosive liquids such as water tanks, waterways, oil tanks, water towers, and deep wells.

SELECTION TABLE			
LT100	Liquid level transmitter		
Range	1	0-0.5M(min)	
	2	0-1M.....500M	
	S	Customer customization	
Cable	1	1M	
	2	0-1M.....500M	
	S	Customer customization	
Output	1	4-20mA	
	2	0-5V	
	3	0-10V	
	4	RS485	
	S	Customer customization	
Power supply	1	12VDC	
	2	24VDC	
	S	Customer customization	

Example: LT100-1-1-1-1

Liquid level transmitter, Range 0-10M, Cable 20M, Output 4-20mA, Power supply 12VDC

Parameters

Measuring medium: water, oil, liquid (compatible with SS316L)

Cable length: 1-500M (choose as needed)

Output: 4-20mA, 0-5V, 1-5V, 0.4V-4.5V, 0-10V, RS485

Power supply: 12V/24VDC, 12V-24VDC, customized

Accuracy: 0.2%F.S, 0.5% F.S. (default)

Overload capacity: 150% F.S.

Response time: ≤10ms

Protection grade: IP68

Long-term stability: ±0.2%F.S./1year

Medium temperature: -20°C-85°C

Operating temperature: -10°C-65°C

Temperature drift: 0.05%F.S./°C(-10°C-65°C)

Features

- 1.Adopting imported diffused silicon sensors, the measurement accuracy is high and the performance is stable.
- 2.High protection level, multiple protections, three sealing processes, making use more reassuring.
- 3.Integrated structure sealing, waterproof, dustproof.
- 4.304 stainless steel shell, sturdy, reliable, and corrosion-resistant.
- 5.Polyester ammonia cable with internal air duct, oil resistant and waterproof, improving signal transmission.

Working Principle

When the sensor is inserted into the measured liquid and reaches a certain depth. The pressure of the measured medium is introduced into the positive pressure chamber of the sensor, and the atmospheric pressure on the liquid surface is connected to the negative pressure chamber of the sensor through a cable air duct to counteract the pressure exerted by the sensor. The sensor measures the pressure and establishes a linear correspondence between the output signal and the liquid depth to achieve liquid depth measurement.

Product Material

Cable: Polyurethane

Sealing cap: ABS engineering plastic

Shell: 304 stainless steel (Customizable SS316)

Diaphragm: 316L stainless steel

Electrical Protection

Short circuit protection: permanent

Reverse protection: No damage, but not working

Insulation resistance: ≥ 100MΩ 500VDC

Insulation strength: 500VAC

Environmental Condition

This product has a protection level of IP68. When installing and using it, pay attention to waterproofing. The measurement should be for non corrosive liquids.

Environmental temperature: -20°C-85°C

Operating temperature: -10°C-65°C

Environmental humidity: 0-95% RH (no condensation, no dew)

Wiring Diagram

